AI and Biomedicine: An STS Perspective

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Abstract

While the future of biomedical artificial intelligence (AI) is promising, it faces significant challenges that are revealed through science and technology studies (STS). Biases in training data, a cycle of performativity in its development, patient distrust, and the need to establish a responsible integration timeline all stand in the way of the huge potential of AI in healthcare. By carefully evaluating and correcting biases in AI training data, breaking the cycle of performativity, building patient trust, and determining a responsible timeline for AI integration, we can harness AI to revolutionize health care, enhancing patient outcomes in a new era of personalized medicine.

Introduction

The rise of artificial intelligence (AI) is steadily reshaping the world as we know it. From simplifying every-day tasks[1] to solving complex scientific problems[2], AI is opening new avenues of convenience and innovation. The recent advent of large language models (LLMs) like ChatGPT has brought about immense hype as they are capable of answering complex questions, providing detailed summaries, translating different languages, and writing reliable code, often at a level indistinguishable (or better than) humans[3–8]. Recent headlines boast that these models pass prestigious, graduate-level exams, including the Bar[9] and USMLE[10], even with training that is not tailored to these fields[5]; and that they are now beginning to show "sparks of artificial general intelligence" including possessing mathematical capabilities, interacting with the world, interacting with humans, being influenced by society, and exhibiting discrimination[11]. These accomplishments, among others, are quite remarkable, which has led to immense hype surrounding the integration of AI into various fields[12–14], including biomedicine[15].

In essence, AI models are tools that predict outcomes based on a statistical algorithm that processes some input data. This input data should provide information that correlates to the predicted output[16]. For example, it would be useful to use chest x-rays as an input to an AI model that is designed to predict whether someone has pneumonia[17]. By training these models on large collections of input data with known outcomes, they learn how the input data influences the outcome, refining their predictive capabilities[18].

The rise of the electronic health record (EHR) in recent years has led to an unprecedented accumulation of patient data, which serves as an invaluable resource for medical researchers, especially those interested in the intersection of health and AI[19,20]. This information, containing diverse patient demographics, histories, diagnoses, and vital signs[21], serves as the foundation for AI's integration into medical practice[22]. Using this rich collection of data, AI promises to revolutionize the practice of medicine, reducing costs and elevating standards of care[23,24].

While the future of medical AI is bright, its grand promises will not come without overcoming major obstacles revealed by analysis through the lens of science and technology studies (STS) including biases in training data, a performative cycle of development expectations, patient distrust, and the challenge of integrating AI responsibly and timely. Tackling these challenges is key to developing reliable and equitable biomedical AI and may be accomplished through an interdisciplinary approach to research, continuous physician education, and effective communication with patients.

Theory and Methods

Documents

Documents are more than just static artifacts that provide a repository of past information; they are active participants in the societies they represent, especially in the context of STS[25]. This dynamic nature is particularly apparent in the digital transformation of documents such as the shift from paper-based medical documentation to EHRs. The rise of EHRs exemplifies how documents evolve from mere data repositories to integral components of social processes.

Considering EHRs and their relevance to the integration of AI in biomedicine, it is essential to ask: What kind of data do EHRs contain, and how can this data both aid and hinder the progress of medical AI? How do socioeconomic factors impact the evolution of the EHR, and how does this have implications for the development of reliable and equitable medical AI?

Sociology of Expectations

The sociology of expectations is a key concept in understanding the dynamics of science and technology development. This approach emphasizes that expectations, visions, and imaginaries are more than just passive descriptions of potential outcomes; rather, they play an active role in shaping and influencing the trajectory of scientific and technological progress. The sociology of expectations suggests that these anticipatory elements guide, mobilize, and coordinate the activities of various stakeholders including scientists, innovators, policymakers, regulators, NGOs, and societal actors[26]. In the intersection of AI and biomedicine, the sociology of expectations offers an important lens for analysis. Using this approach, it becomes evident that promises and visions surrounding AI in biomedicine have a performative effect, driving research and development in specific directions. It is crucial to examine what these expectations entail and how they manifest in biomedical AI: What are the visions and imaginaries surrounding AI in this context? How might these expectations influence the actual development of AI technologies and biomedical applications?

Responsible Research and Innovation

Responsible research and innovation (RRI), a concept that has gained significant traction in the field of STS, emphasizes the importance of anticipating, reflecting, and responding to the ethical, social, and environmental implications of scientific and technological advancements[27]. This approach is crucial in the realm of medical AI, especially considering the hype surrounding the field and the use of EHRs that reflect current societal biases. The rapid development and deployment of AI technologies, often spurred by hype and market demands, may overlook the thorough testing and ethical considerations typical of traditional medical research[28]. With this in mind, it is important to examine: How do expectations surrounding medical AI have implications in RRI? Should AI follow the same research and development timeline as traditional medical research? How can AI be integrated in an effective and timely manner, while still being conscientious of ethical implications, data privacy, and equitable access?

The Black Box

RRI places a strong emphasis on the concept of the "black box," which reflects the often opaque and inscrutable nature of modern technology[27], including medical AI. This "black box" nature poses significant challenges in terms of data privacy, equal access, and responsible implementation of AI in healthcare[29]. In order to overcome these challenges, we must ask: How does the concept of the "black box" play a role in AI, and what are the implications? How can we ensure the transparency and accountability of AI systems in healthcare? How can AI be integrated into the practice of medicine in a way that respects and enhances human understanding and decision-making, rather than undermining it?

Literature Review

The Electronic Health Record

The use of medical records dates back to 1600 BCE, when Egyptian physicians recorded surgical cases on paper made of papyrus[30]. Medical records, in the modern sense, were developed in the early 1900s, where patient data was recorded and organized in a standardized format[31]. The first EHR was created in 1972 by the Regenstreif Institute in the United States, which was considered a major breakthrough but was not readily adopted, mainly due to concerns regarding financial burden[32]. Today, EHRs play an integral role in medicine across the United States, with 96% of hospitals[33] and 88% of private practices[34] adopting them. This widespread adoption can be attributed to the Health Information Technology for Economic and Clinical Health (HITECH) Act, via which the US government has provided over $30 billion in financial incentives for the adoption of EHRs[35]. In modern practice, EHRs are used to document everything in medicine, and they contain a vast array of information, outlined in **Table 1**.

Table 1. Common and emerging data types found in EHRs[21]. Starred data types indicate data used in the Obermeyer study. (Adapted from Ehrenstein et al., 2019)

DATA TYPE	EXAMPLE
Demographics *	age, sex/gender, race
Diagnoses*	diagnosis, severity, medical history
Problem List	active diagnosis, resolved diagnosis
Family History	familial disorders, risk factors
Allergies	food and medication allergy, anaphylaxis
Immunization	DTaP, HepB, IPV
Medications*	Prescriptions written
Procedures*	inpatient, outpatient
Lab Orders/Values	CBC results, HbA1C levels
Vital Signs	BMI (weight and height), blood pressure
Reports	radiology, pathology and other reports
Utilization*	cost, hospitalization
Emerging Data Types	
Biosample Data	meta data about a biological sample
Genetic Information	genome sequence data
Social Data	income level, education, employment status
Patient-Generated	mHealth, patient communications
Community	community specifications
Geo-spatial	neighborhood built environment
Surveys	HRA, PHQ9, PROs
Free Text	various markers (e.g., specific test results)
Other Data Types	clinical workflow data

The EHR is more than a static repository of patient information; instead, it is constantly evolving[36], much like the patients it represents. Patients' dynamic health statuses and medical histories are in constant flux due to physiological changes, lifestyle choices, and encounters with the healthcare system, all of which may be accounted for in the EHR[21]. The EHR's fluidity also captures broader societal trends including the advancement of medical practice, innovations and diagnostic tools, and the growing understanding of how psychosocial elements can impact patient health and behavior[20,37]. On top of this, the type of information contained within the EHR is constantly evolving, with new data types, as shown in Table 1, being integrated over time[21]. Harnessing this rich collection of patient information, AI promises to revolutionize the medical field – efficiently analyzing and transcribing medical documents, aiding in prognosis and diagnosis, and potentially elevating standards of care for patients[19,38].

In healthcare, there are already various care-management algorithms (CMAs) that are integrated into EHRs that are designed to target high-risk patients for enrollment in care-management programs. These algorithms are designed to help physicians identify patients who will reap the greatest benefit from this extra care (i.e. the sickest patients) and are applied to roughly 200 million patients in the US annually[39].

While there are certainly benefits to this kind of technology, especially in the realm of preventative healthcare, there are also drawbacks. For instance, there is a growing concern that AI may perpetuate racial and gender biases, either via biases of individuals building them or via the data used to train and evaluate them[39]. It has already been shown that AI facial detection software performs worse on recognizing the faces of women and Black individuals, compared to men and White individuals, due to their predominant training on white men[40].

Recently, Obermeyer at al. (2019) investigated whether differences exist in CMA recommendations for White and Black patients. They chose an algorithm for which a wide variety of data was available, including inputs and outputs of the model, as well as patient outcomes, for a total of 49,618 cases. The algorithm they chose works based on a variety of inputs, namely demographic features (excluding race), diagnoses, procedures performed, medications, and detailed cost information (see **Table 1**). These inputs are fed into the model, which calculates a risk score based on expected future healthcare expenditure[39].

The researchers found that there is no significant difference in the total future costs between Black and White patients at the same calculated risk score, meaning the algorithm was able to accurately predict future expenditure. However, they found that, on average, White patients have significantly fewer chronic conditions than Black patients at the same calculated risk score do (see **Figure 1**). Moreover, they found that the characteristic biomarkers for various chronic conditions, including hypertension, diabetes, hypercholesterolemia, renal failure, and anemia, indicated significantly higher severity of disease in Black patients when compared to White patients at the same algorithm risk score. These findings indicate that, because of the way the algorithm was designed and implemented, Black patients were less likely to receive additional care measures than their equally sick White counterparts[39].

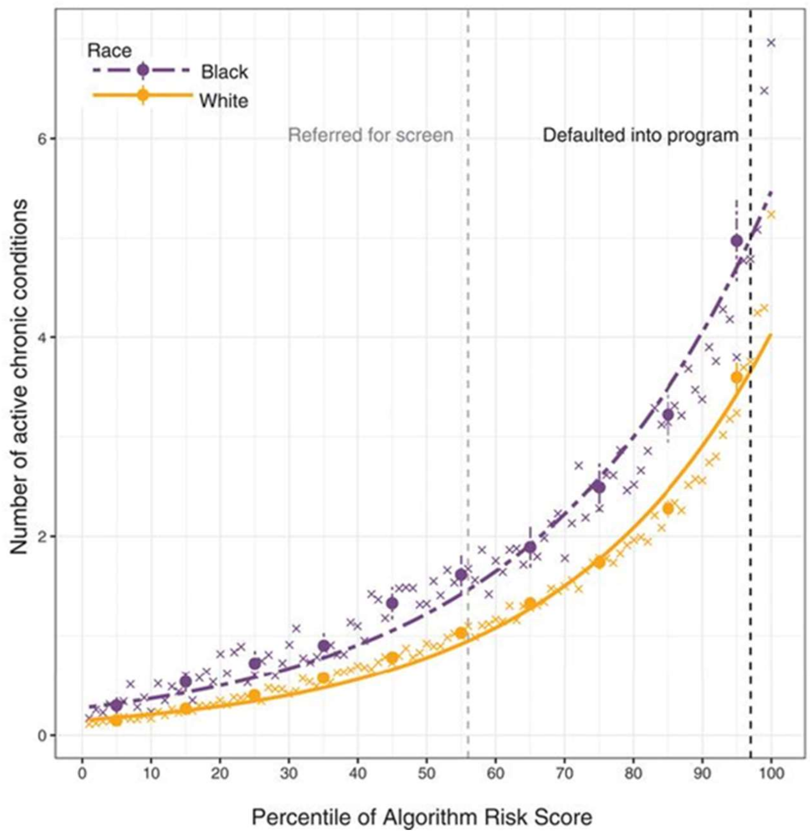

Figure 1. Number of active chronic conditions versus algorithm risk score[39]. The grey dashed line indicates the threshold at which patients are referred for further care; the black dashed line indicates the threshold at which patients are automatically enrolled in further care. (Taken from Obermeyer et al., 2019)

They found that these disparities ultimately arise because the model uses future costs as a measure of health. They argue that poorer individuals face substantial barriers to healthcare, and that since there is systemic inequality that negatively impacts the Black population, Black patients are less likely to access and spend money on healthcare than White patients. They further note that "taste-based discrimination" plays a role in this disparity, with Black patients potentially receiving fewer services or less aggressive treatment options than their White counterparts. The authors also touch on the doctor-patient relationship, noting that Black patients have less trust in the healthcare system than White patients, due to a multitude of reasons. Overall, they suggest that collective effect of these factors leads to decreased average healthcare spending by Black patients compared to equally ill White patients, and since the model was trained to equate illness with future spending, it propagated this bias[39].

Expectations and Imaginaries of Medical AI

Amidst all the recent AI hype, researchers have been examining the expectations and imaginaries surrounding the integration of AI into the practice of biomedicine, especially those of physicians. A survey of 791 psychiatrists across the US found that only 4% of respondents believed that AI would fully replace them, while 75% believed that AI could improve medical documentation and

54% believed that AI could synthesize information to establish prognoses and diagnoses[41]. A survey of 1032 radiologists across Italy revealed that 73% believed AI would lead to a lower diagnostic error rate and 68% believed that it would optimize their workload[42]. In this survey, 89% of respondents were not afraid of losing their job to AI. A survey of 487 pathologists across 54 countries found that 75% believed that AI would bring about improvements to efficiency and accuracy[43]. A survey of 303 physicians throughout Germany found that 90% believed that humans and AI will work side-by-side in the future, and 92% believed that AI would be able to improve patient care by identifying drug interactions[44]. Another survey of neurosurgeons revealed that 60% reported already using AI to predict outcomes pre-operatively[45]. A survey of 720 UK general practitioners found that respondents believed AI would have benefits in reducing documentation work, but that it would have limitations due to social and ethical concerns[46]. A survey of 112 physicians in a teaching hospital in Malaysia revealed that physicians who worked in a non-clinical setting or who were more tech-savvy were more likely to have favorable attitudes towards AI[47].

A recent study aimed to investigate which specific aspects of medicine physicians believed AI would be most beneficial in. The results of this study are shown in Figure 2. The question researchers were most interested in was "What is the likelihood that AI will replace physicians in performing tasks as well as or better than the average physician?" They found that physicians seemed to believe that AI would be most likely to be as good or better at synthesizing patient data to determine prognoses (72%) and to reach diagnoses (69%). These results seem to support the other work, which also suggests that physicians believe that AI will allow for better integration of information, and for establishing prognoses and diagnoses[48].

Aside from physician perspectives, there have also been recent studies looking into patient attitudes. A survey of 11,004 patients across the United States revealed that 60% would feel uncomfortable if their health care provider relied on AI in their care, and that only 38% believed that the integration of AI into medical practice would lead to better health outcomes. Moreover, they found that 75% believed that providers will "move too fast using this technology, before fully understanding the risks for patients." Increasing comfort levels and increasing confidence levels seemed to directly correlate with decreasing age, increasing income, and increasing level of education. Interestingly, 70% of respondents believed that bias is a problem in American healthcare, and of those respondents, 51% believed that AI would be helpful in reducing this bias, 33% believed it wouldn't make a difference, and 15% believed it would make it worse. Asian respondents were most favorable, followed by White, Hispanic, and Black respondents, who were the least favorable. In another study which sampled 2,675 patients and oversampled minority populations, patients were found to be more favorable towards AI when provided with accurate information about AI's effectiveness in medical practice and/or when their primary care physician (PCP) recommended the use of AI in their care.

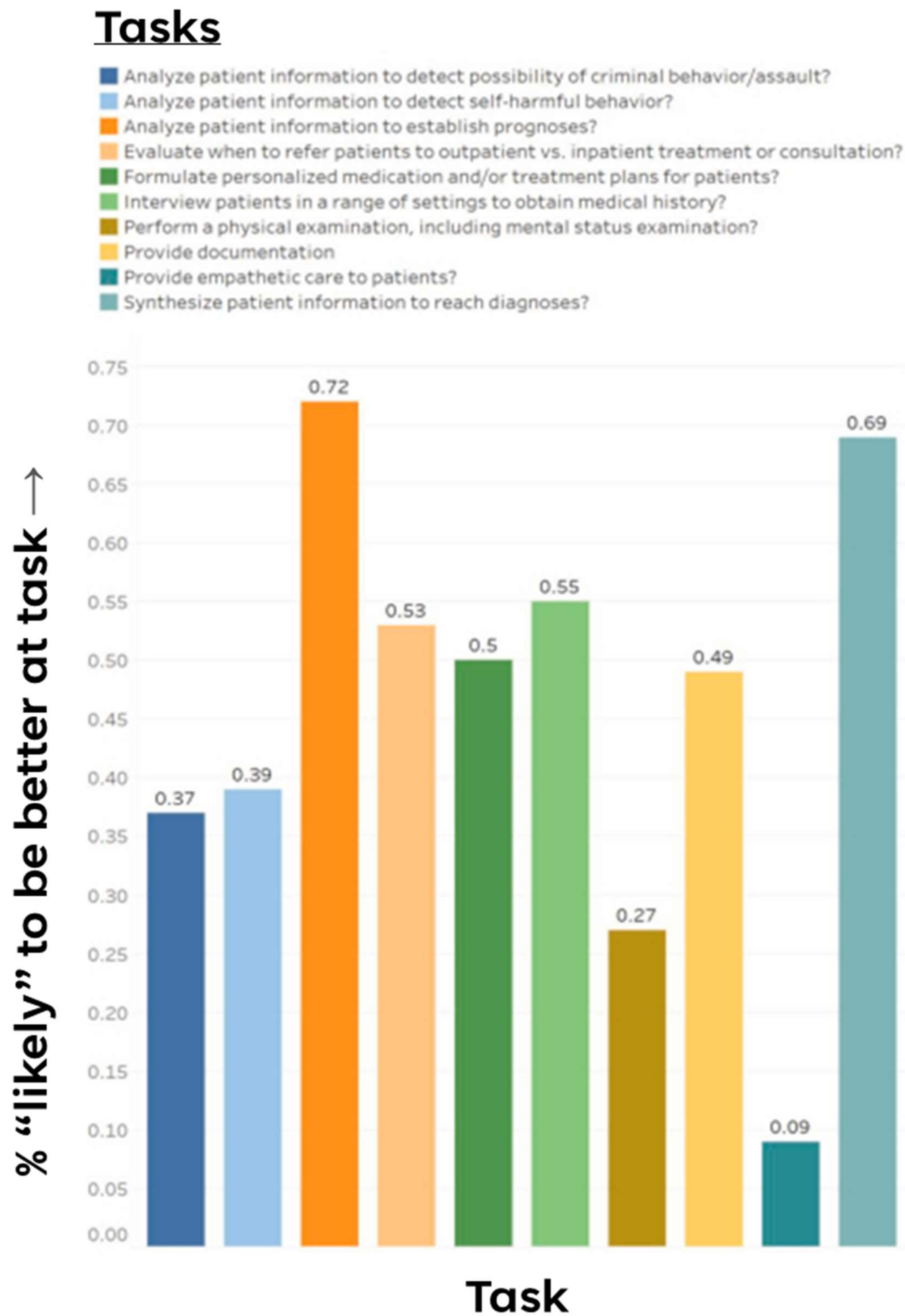

Figure 2. Fraction of 114 physician respondents who answered "Likely" to the question "What is the likelihood that AI will replace physicians in performing tasks as well as or better than the average physician?"[48] (Adapted from Al-Medfa et al., 2023)

Responsible Research and Innovation

With the development of modern technology, including AI, there has historically been a "move fast and break things" mindset[49]. This is often driven by a desire for innovation, with researchers and developers embodying the belief that taking on greater risks will lead to greater rewards[50]. While this can be true, rapid technological development, especially in the realm of AI, can be disruptive to society and inconsiderate of broader impacts. Recognizing these challenges, the current trend is shifting towards a more responsible approach to technological innovation[51].

The traditional timeline of development in medical technology, like in drug development or medical equipment deployment, has been marked by a rigorous and methodical process. This often includes extensive research, multiple phases of clinical trials, and thorough regulatory review to ensure safety and efficacy. This cautious approach, though time-consuming, is essential due to the direct impact these technologies have on human health and society[52,53].

In contrast, many researchers interested in the introduction of AI into healthcare advocate for a more streamlined process. For example, Dr. Parul Friedman and Dave McMullin, two researchers exploring the capabilities of AI in cardiovascular diagnostics, shared their thoughts in an interview. They coined the phrase "move fast and cure things," which represents their mindset that AI will accelerate the detection and diagnosis of medical conditions, like heart failure, allowing for quicker, and potentially more cost-effective, solutions than traditional methods. While they do advocate for a quicker timeline of development than traditional medical technology, they still underscore the importance of balancing the urgency of innovation with the importance of accuracy and reliability in medical applications[28].

The integration of AI into biomedicine also brings concerns regarding the "black box" nature of AI algorithms. Often times, AI models can produce accurate results, but are unable to provide explanations for their conclusions, which raises questions of accountability and trust. This can be especially problematic when clinicians need to convey their reasoning behind a particular treatment or diagnosis[29].

Discussion

EHR Data: A Double-Edged Sword

EHRs represent a fundamental shift in the way patient data is collected, stored, and utilized in the practice of medicine. They are repositories of vast and diverse patient data, offering a foundation for developing AI that can transform healthcare delivery. AI algorithms have already proven effective at analyzing patterns across patient records to identify which ones might benefit most from additional care. On top of this, the wealth of information contained in the EHR may be able to facilitate training of AI models that will advance personalized medicine. By relying on AI algorithms that use specific patient data as input, physicians will be able to provide care and tailor treatments to individual patients based on their unique medical histories and genetic makeup, rather than basing care on averages.

However, the utility of EHR data in the realm of medical AI is a double-edged sword. A significant challenge of training medical AI on EHR data is the fact that this data is not a neutral entity, but rather a reflection of existing healthcare practices, patient demographics, and social structures. Because of this, inherent biases within the health care system, whether related to race, gender, or socioeconomic status, are likely to be mirrored in the EHR data. Subsequently, when AI algorithms are trained on this data, biases may be perpetuated, or even amplified, when they are deployed in medical practice.

The case of racial disparities in CMAs is a glaring example of this issue. The algorithm examined in the Obermeyer study, designed to predict future healthcare costs as a proxy for health needs, failed to account for the socioeconomic factors that lead to disparities in healthcare access and spending, leading to its underestimation of health needs for Black individuals and likely perpetuating an existing bias in healthcare. In this case, the bias was accounted for by simply adjusting what the algorithm used as a measure of health, revealing how careful consideration of the data used in the training of medical AI is necessary in the design of equitable models.

Visions: The Cycle of Development and Expectations

Examining physician attitudes and expectations for AI in medical practice is crucial, as they are the ones who will be using it in the clinical setting, and their visions will ultimately influence the way it is integrated. Physician attitudes towards AI, as evidenced by various surveys and studies, are generally positive, with a seemingly strong belief in AI's potential to improve medical documentation, diagnosis, and prognosis. This optimism among physicians, among hype surrounding modern technology and AI in general, likely contributed to the development, and arguably rapid deployment, of biased CMAs.

Interestingly, there seems to be a performative effect of physician expectations for AI. Types of AI applications that physicians anticipate being beneficial, such as aiding in prognosis and diagnosis, are indeed the ones being actively developed, implemented, and refined over time. For instance, CMAs are essentially predicting prognoses of patients, and according to **Figure 2**, physicians expect AI to be best at this task. Furthermore, it seems that there is a cycle where expectations shape development, and development reinforces expectations. Given this cycle, it would be interesting to investigate the potential of AI algorithms in areas where physician expectations are currently low or uncertain. For instance, according to **Figure 2** and Doraiswamy et al. (2020), a relatively low percentage of physicians anticipate their jobs to be entirely replaced by AI[41,48] , especially in the realm of "providing empathetic care." While it is unlikely that an AI could truly provide empathetic care, it is interesting to consider whether other aspects of care that physicians do not have high expectations for, including detecting self-harm or performing a mental status examination (see **Figure 2**), could be well-suited for AI. It is possible that these expectations are comparatively lower than establishing prognoses and diagnoses based on physician visions and imaginaries for AI. That is, physicians are less likely to envision (or want) AI performing mental status examinations (or providing empathetic care), so there is less willingness to research AI applications in these areas. Moreover, the lack of research on these applications contributes to their low anticipation, while the abundance of research on envisioned applications contributes to their high anticipation. While this is not inherently negative, if AI could excel in tasks traditionally deemed challenging or unsuitable, this could break the cycle of performativity and open new avenues for biomedical AI. However, such algorithms might initially face hurdles in gaining traction, as they are not part of the existing cycle of expectation and development, possibly leading to slower recognition (or none at all) within the medical community.

While consideration of physician attitudes is important, equally, if not more, important are the attitudes of patients, to whom this technology will be applied. Considering that a significant portion of patients feel uneasy about AI becoming integrated into their care, coupled with the fact that patient attitudes towards AI seem to improve when provided with accurate information regarding its use, it is imperative that physicians not only comprehend the technology, but that they are able to effectively communicate their understanding of, and confidence in, it. This involves not only understanding the technical aspects of AI but also being able to explain these in a manner that is accessible and reassuring to patients. Therefore, training and education programs for healthcare providers should be implemented with a focus on both the technological and communicative aspects of AI. If patients are well-informed about the capabilities and limitations of AI, they will be better positioned to engage in shared decision-making with their providers. This process not only empowers patients but also strengthens the trust between them and their providers, which is a cornerstone of effective health care delivery, and is a key element in unlocking the full potential of medical AI. When patients have confidence in the AI tools used by their health care providers, they are more likely to accept and benefit from the advanced insights and efficiency these technologies can offer. Thus, cultivating patient trust and understanding is not just beneficial for individual patient-provider interactions but is essential for the broader acceptance and success of AI in healthcare.

RRI: Moving Fast and Curing Things

AI is rapidly evolving and has the potential to significantly enhance healthcare delivery. However, it primarily serves as a decision-support tool, rather than a direct form of patient intervention, like drugs or surgical equipment. Consequently, the risks associated with AI deployment in health care, although significant, are of a different nature and magnitude than traditional medical technologies. Therefore, the development of medical AI calls for a unique approach that strikes an equilibrium between the “move fast and break things” ethos of Silicon Valley and the slow, meticulous progression of traditional medical technologies.

Adopting a “move fast and cure things” mindset could entail swift development and testing of AI tools in clinical practice, followed by continuous monitoring and cycles of feedback, evaluation, and refinement. An auditing system like this could constantly monitor AI systems for biased outcomes, preventing (or at least reducing) issues like those with CMAs. This developmental model is particularly suited to the dynamic nature of AI, where the implementation of algorithms, along with the algorithms themselves, could learn and improve over time. In such a deployment model, the role of interdisciplinary collaboration becomes pronounced. Ideally, teams comprising AI researchers, physicians, administrators, patients, sociologists, and ethicists should work together to ensure AI tools are developed and refined in a way that aligns with realistic clinical practices, socioethical standards, and patient needs.

The Black Box: A Challenge and an Opportunity

Amidst these considerations, the “black box” nature of AI and healthcare stands out as a significant challenge. This opacity, where the inner workings of AI algorithms are not easily understandable or interpretable, especially in the case of LLMs, poses risks in terms of accountability and trust. Physicians, and by extension, their patients, may find it difficult to comprehend the basis of AI-assisted decisions. This lack of understanding can create barriers in the acceptance and ethical application of AI and healthcare, as well as in effective communication between health care providers and patients.

Addressing the “black box” nature of AI can be approached from two angles: enhancing physician and patient understanding of AI technology and/or improving the technology itself. The former involves comprehensive education and training for health care professionals who will be using AI in their clinical practice. As discussed previously, such programs should not only cover the technical usage of AI but also focus on interpreting and communicating AI decisions in a way that is understandable to patients.

Improving the technology itself involves developing more transparent and interpretable AI systems. Efforts in explainable AI (XAI) aim to make AI decisions more understandable to humans, allowing health care providers to see the reasoning behind AI recommendations[54]. This kind of transparency may be crucial for ensuring that AI supports, rather than undermines, clinical judgment. It may also provide a means to identify and rectify biases within AI systems more reliably and efficiently.

Ultimately, even without XAI, there is still significant potential for improving our understanding of medical AI through rigorous research and testing in clinical environments. For instance, Obermeyer et al. were able to discern and eliminate the bias from the CMA they were using by thoroughly evaluating a current practice. While this kind of monitoring would likely prove more challenging with sophisticated AI systems like LLMs, efforts are already underway to understand these models in fields like psychology and cognitive science, which may offer insight into the black box of current AI[55–59]. As more research comes out, healthcare professionals can gain empirical evidence of AI’s decision-making patterns and outcomes, even if the underlying algorithmic processes remain complex. This empirical understanding may foster increased trust and confidence in AI systems, enabling physicians to integrate AI tools into their practice more effectively.

Overall, while the development of inherently transparent AI systems remains an ideal goal, the practical understanding of AI’s capabilities and limitations in healthcare can still be significantly advanced through continuous research, clinical testing, and collaboration between technological developers and healthcare practitioners. This approach ensures that even non-XAI systems can be used responsibly and effectively in medical settings, enhancing patient care while maintaining the trust and confidence of providers and patients alike.

Conclusion

The integration of AI into healthcare, guided by the wealth of data contained in the EHR, promises to bring about a transformative era in medicine. Considering its potential, it becomes essential to think about the challenges of inherent biases and ethical complexities alongside the promise of personalized care. The "black box" nature of AI remains a significant hurdle, yet it presents an opportunity for continuous learning and adaptation within the healthcare system and within AI research. Efforts towards enhancing understanding and transparency in AI, through interdisciplinary collaboration and iterative research, are critical for fostering trust and ensuring responsible integration. As AI continues to evolve, its integration into healthcare demands a conscientious approach that addresses social considerations, upholds ethical standards, and prioritizes patient-centered care. Ultimately, the journey of AI in healthcare is not just about technological advancement, but about harnessing these innovations in a way that enriches medical practice, contributes to better patient outcomes, and promotes equitable healthcare for all.

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